CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of Bachelor of Technology (B.Tech.) (EE)

University Examination November 2010

EE207 Network Analysis

Date: 26/11/2010, Friday **Time: 10:30 a.m. to 1:30 p.m.** **Maximum Marks: 70**

Instructions:

1. Write both the sections in separate answer books.

2. Take assumptions if required and mention them.

Section –I

			Marks
Q.1	(a)	State Thevenin's Theorem. Determine Thevenin's equivalent circuit across terminals AB shown in figure 1 (a).	5
	(b)	Determine current flowing through 5Ω resistance in the circuit shown in figure 1 (b) using Norton's Theorem.	5
Q.2	(a)	Determine the maximum power delivered to the load in the circuit shown in figure 2.	5
	(b)	For the graph shown in fig 3 write the tie-set schedule and obtain the relation between branch currents and link currents.	5
		OR	
Q.2	(a)	State Norton's theorem. Determine Norton's equivalent circuit for the circuit shown in figure 4.	5
	(b)	For the graph given in fig.5 write the cut-set schedule and obtain the relation between tree branch voltages and branch voltages.	5
Q.3	(a)	Find z-parameters for the circuit shown in figure 6.	5
	(b)	Find ABCD parameters in terms of z parameters for a generalized 2-port network.	5
	(c)	Find y parameters in terms of z parameters for a generalized 2-port network.	5
		OR	
Q.3	(a)	For a general 2-port network find z parameters in terms of y parameters.	5
	(b)	Find transmission parameters for a general 2-port network.	5

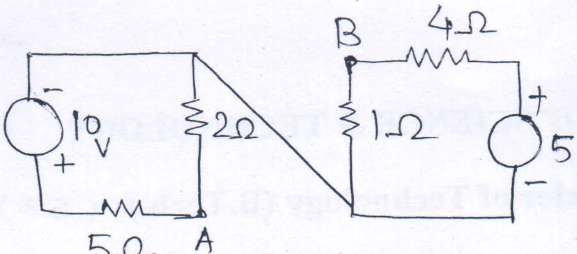

fig 1(a)

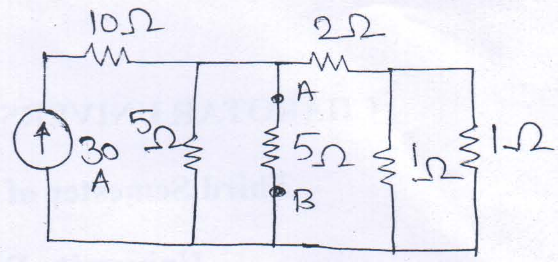

fig 1(b)

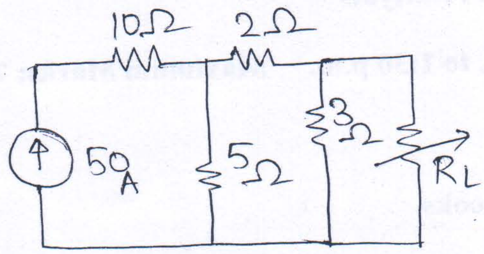

fig 2

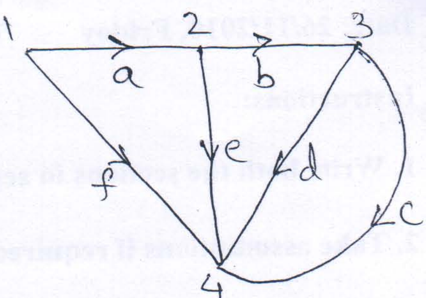

fig 3

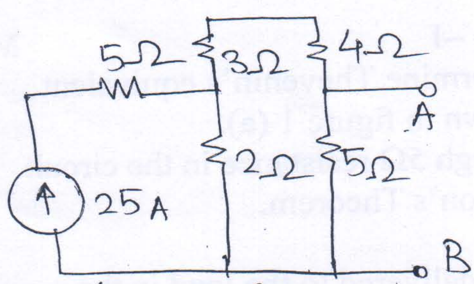

fig 4

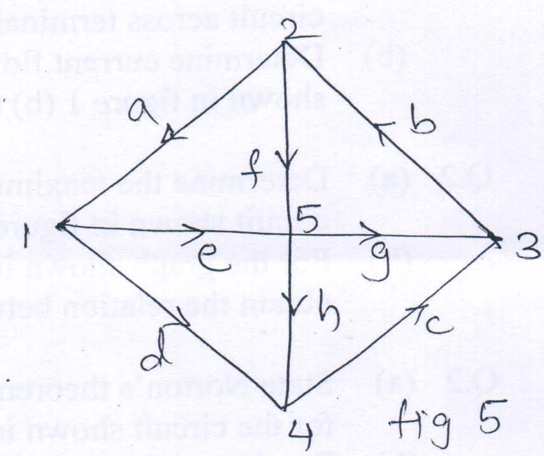

fig 5

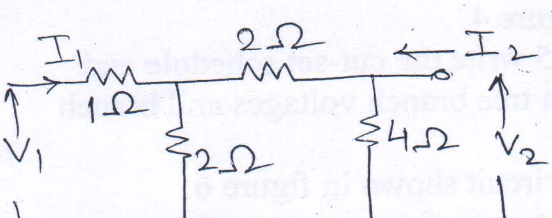

fig 6.

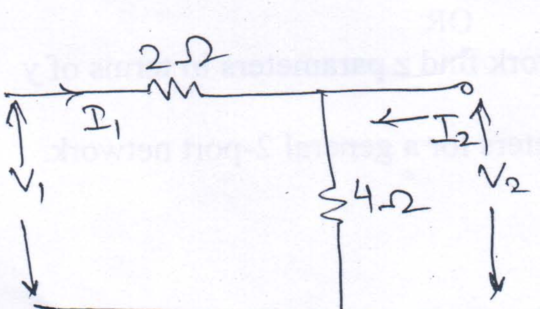

fig 7.

(c) Find h-parameters for the circuit shown in figure 7. 5

Section-II

Q.4 (a) Write I_{out} equation in terms of circuit parameters for the figure 8. Find Current gain if $R_2 = R_{in}$. 5

(b) Define GATE function and write sample expression of it. 5

OR

Q.4 (a) Find equivalent values of L and C for the circuit shown in figure 9. 5

(b) Define initial and final value theorem. 5

Q.5 (a) Find current through 60 Ω resistance shown in figure 10 using Nodal Analysis. 7

(b) Find Vo(t) using Laplace Transform for the circuit of figure 11. 8

OR

Q.5 (a) Find current through 60 Ω resistance shown in figure 10 using Mesh Analysis. 7

(b) Find loop current expression with no initial conditions for the circuit shown in figure 12 with V1= 5 u(t-1) volts. 8

Q.6 (a) Draw unit impulse, unit step and unit ramp functions delayed by 'T' interval from origin. Write expression of each function. 5

(b) Write waveform expression for the wave shape shown in figure 13 and find laplace expression of the same. 5

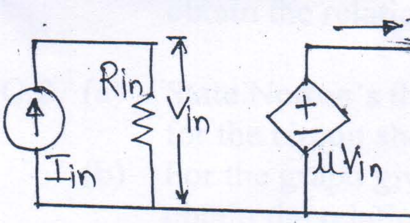

fig.8

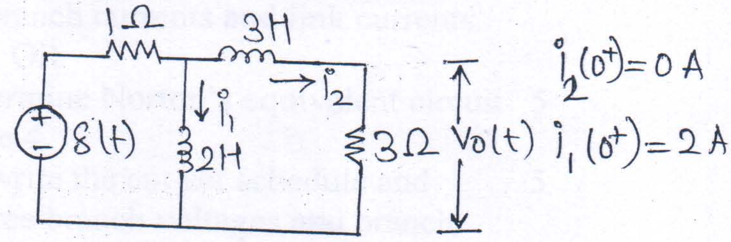

fig 11

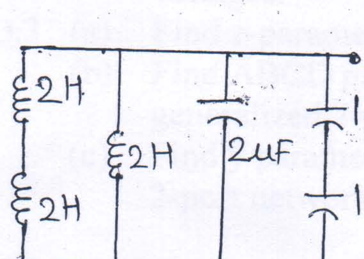

fig.9

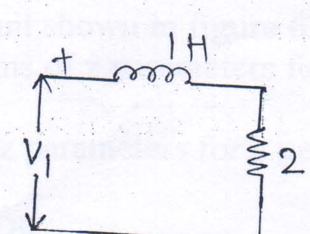

fig 12

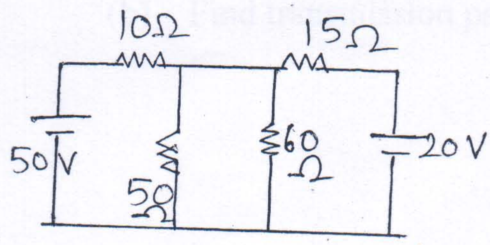

fig 10

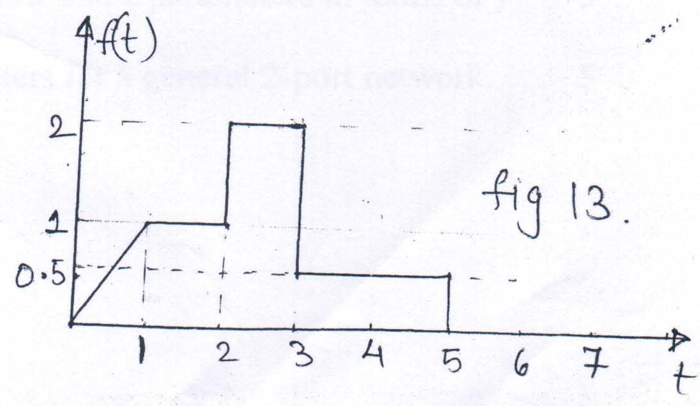

fig 13.